ARMORER'S BENCH TECHNICAL MANUAL

Lewis Automatic Machine Gun

Savage Arms / BSA (Birmingham Small Arms) • Model 1914 / 1917 / 1918 (.303 British & .30-06 Springfield)

Manufacturer:	Savage Arms / BSA (Birmingham Small Arms)	**Category:**	Machine Guns
Action Type:	Gas-Operated Machine Gun (Open Bolt)	**Era / Service:**	Historic Military (WWI / WWII)
Calibers:	.303 British (Mk VII), .30-06 Springfield (US M1917)	**Covered Models:**	Model 1914 (BSA .303), Model 1917 (Savage Arms .30-06), Model 1918 Aircraft (Stripped Radiator)

KEY ARMORER TASKS & BENCH OPERATIONS

Core Procedures: Return Mainspring Tension Calibration (12–14 lbs) • Striker (Firing Pin) Protrusion Check (0.050"–0.060") • Gas Regulator Cup Rotation ('S' / 'L') • Field Stripping & Rack Timing • 47-Round Pan Magazine Servicing

CRITICAL ARMORER TOLERANCES & SPECIFICATIONS

SPECIFICATION / CLEARANCE	ARMORER TOLERANCE & BENCH SPECIFICATION
CALIBERS	.303 British (Mk VII rimmed) / .30-06 Springfield (US M1917/1918 rimless)
OPERATION	Gas-operated (cup piston), open-bolt, rotating 4-lug bolt, rack & pinion clock spring return
FEED	47-round pan magazine (infantry) / 97-round pan magazine (aircraft/naval)
CYCLIC RATE	500 - 600 rounds/min
SPRING TENSION	12 to 14 lbs pull measured at cocking handle (approx 2 to 2.5 full turns of gear casing from slack)
STRIKER PROTRUSION	0.050" to 0.060" through bolt face at full battery
GAS REGULATOR	Two-orifice cup: 'S' (Small - normal operation) / 'L' (Large - foul/cold conditions)
HEADSPACE 303	Rimmed: GO = 0.058", NO-GO = 0.064", FIELD = 0.070"
COOLING SYSTEM	Aluminum longitudinal fluted radiator enclosed in steel jacket with muzzle blast venturi cowl

IMMEDIATE ACTION & MALFUNCTION DIAGNOSTIC MATRIX

MALFUNCTION CONDITION	PROBABLE CAUSES	ARMORER BENCH REMEDY
Position 1: Cocking Handle Fully Forward **Bolt closed on chamber; gun fails to fire upon trigger pull**	• Empty magazine or pan center rotor jammed. • Broken or shortened striker (firing pin tip chipped or deformed). • Defective primer or dud cartridge. • Piston post striker not reaching primer due to incomplete bolt rotation into battery.	**Immediate Action: Cock gun, observe feedway, cycle new round. If persistent: Check striker protrusion (spec 0.050"-0.060") and examine pan feed pawls.**

Position 2: Cocking Handle at 1" to 2" Mid-Stroke **Bolt stopped during forward travel before fully locking**	• Damaged or dented cartridge case jammed in feedway. • Weak return mainspring (pinion tension under 12 lbs). • Severe carbon encrustation in gas cylinder or piston head. • Weak or deformed cartridge guide spring in feed cover.	**Immediate Action: Pull cocking handle rearward, clear jammed cartridge, check pan teeth. Gunsmith Action: Increase clock spring tension 1-2 clicks; scrape carbon from gas regulator cup and cylinder.**
Position 3: Cocking Handle Near Full Rear **Bolt stopped near recoil limit; failure to extract or eject spent case**	• Broken extractor claw or fatigued extractor spring. • Broken ejector arm (pivoted in left receiver wall). • Ruptured case neck or head separation left inside chamber. • Gas regulator port set too small for dirty conditions (short recoil).	**Gunsmith Action: Inspect extractors and ejector tip. Use broken-shell extractor for separated cases. Rotate gas regulator from 'S' (Small) to 'L' (Large) port.**
Position 4: Cocking Handle Jammed Full Rear **Gun locked open; cocking handle cannot be pushed forward**	• Broken clock return spring binding inside circular gear casing. • Foreign debris or sheared gear tooth jammed in pinion teeth. • Bent piston rod from excessive gas pressure or barrel shroud drop. • Loose set screw or damaged rack.	**Gunsmith Action: Remove buttstock and ease gear casing. Inspect spiral mainspring and pinion teeth. Roll piston rod on precision flat to verify straightness.**

CURATED BENCH VIDEO REFERENCES

PROCEDURE TITLE	INSTRUCTOR / CREATOR	TECHNICAL PROCEDURE SUMMARY
Anvil 033: Lewis Gun Firing Pin and Oprod Repair	Mark Novak / Anvil Gunsmithing	Master-level gunsmithing repair. Machining a replacement striker/firing pin, annealing, hardening to ~50-52 HRC, fitting to op-rod post, and testing bolt lockup headspace.
Project Lightening Ep. 04: Field Strip	C&Rsenal; & Forgotten Weapons	Step-by-step takedown: Buttstock removal, releasing return spring tension safely, dropping trigger group, sliding out bolt & piston, and removing feed arm.